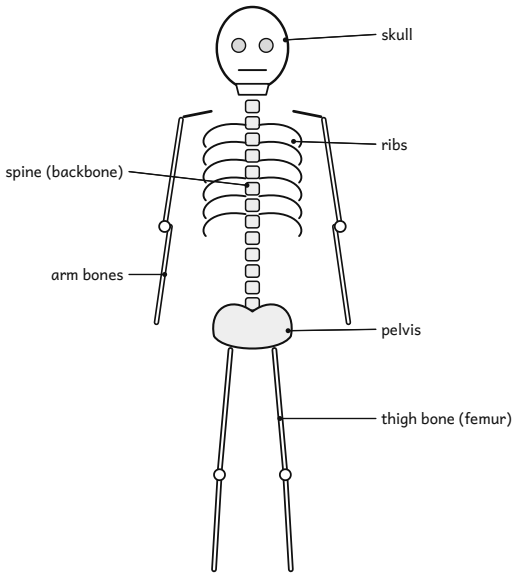
LIVING THINGS

Skeletons, Muscles and Medicines

How skeletons support and protect us, how muscles work in pairs, and using medicines safely



In this pack

- **Learn it:** Skeletons, muscles and medicines ★☆☆
- **Fact check:** Bones and muscles ★☆☆
- **Think it through:** How do we move? ★★☆
- **Investigate:** Do longer arms throw further? ★★☆
- **Real-world case study:** Strong bones from small fish ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science: Stage 3 Unit 2 Looking after ourselves (2.1–2.4).

This pack belongs to: _____ **Class:** _____

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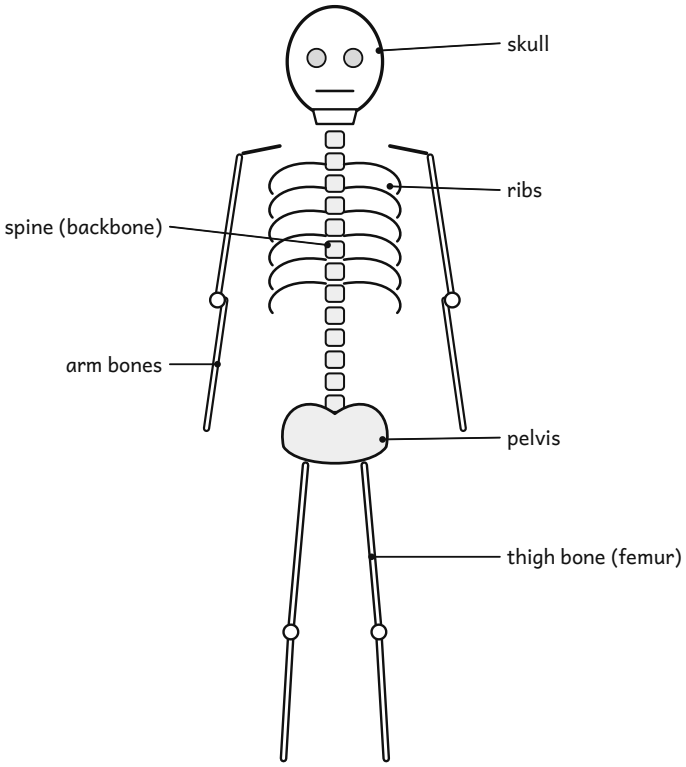
Skeletons, muscles and medicines

Name: **Class:** **Date:**

Learning goals

- Explain how skeletons support and protect the body and grow as we grow.
- Explain how muscles contract in pairs to move bones.
- Explain what medicines are and how to keep the body healthy and safe.

Without a skeleton you would flop like a jellyfish! Humans and many animals, such as dogs, birds, fish and frogs, have a **skeleton of bones** inside their bodies.



The human skeleton.

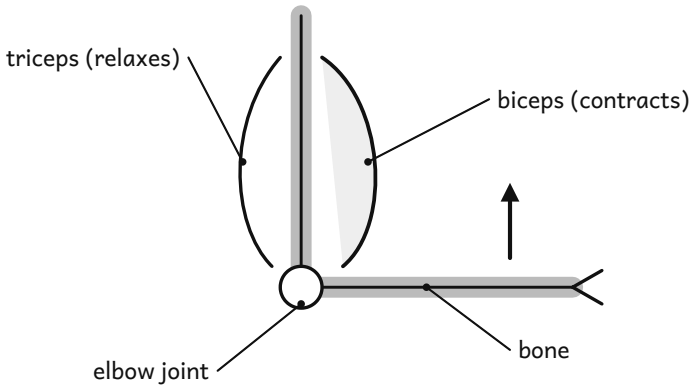
What the skeleton does

- **Supports** the body and gives it shape.
- **Protects** soft organs: the skull protects the brain; the ribs protect the heart and lungs; the backbone protects the spinal cord.
- Helps us **move**: muscles are attached to bones.

A new baby has about 300 small bones. As you grow, some join together, so adults have **206** bones. Bones grow longer and stronger as you grow, so you get taller.

Muscles move bones

Muscles are joined to bones. A muscle can only **pull**. When it **contracts** (gets shorter and fatter) it pulls the bone. Muscles work in **pairs**: to bend your arm, the **biceps** contracts and the **triceps** relaxes. To straighten it, the triceps contracts and the biceps relaxes.



Muscles work in pairs.

Medicines

Drugs are substances that change how the body works. **Medicines** are drugs that help us when we are ill: they can stop pain, fight infections or help the body heal. But medicines can be harmful if they are used wrongly.

> ⚠ **Stay safe**
>
> Only take medicine given by a parent, carer or doctor. Take the right amount at the right time. Never share medicines. Never take medicine that looks like sweets without asking an adult.

Keeping healthy and safe

 **eat well**	 **milk and small fish for strong bones**	 **exercise**	 **sleep**	 **wash hands**	 **wear a helmet**

> 🕮 **Key words**
>
> **skeleton** the frame of bones inside the body
> **support** hold up
> **protect** keep safe
> **contract** get shorter and pull
> **relax** get longer and stop pulling
> **medicine** a drug that helps when we are ill

FACT CHECK · A1 · SKELETONS AND MEDICINES

Bones and muscles

★☆☆
Grade 3

Name: _____ **Class:** _____ **Date:** _____

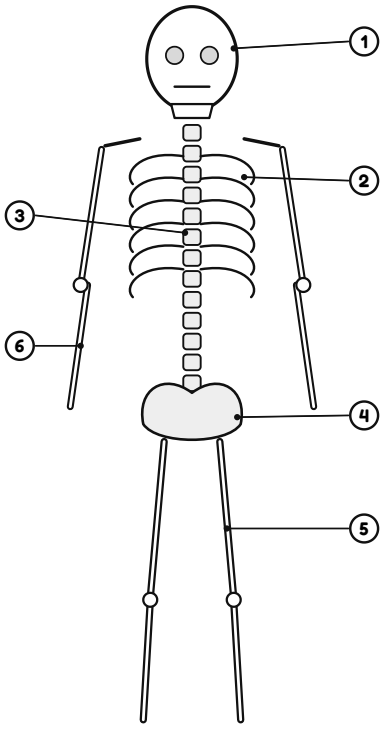
Learning goals

- Label the skeleton.
- Match bones to jobs.

1 Label the skeleton. [6]

LABELS

spine (backbone) **arm bones** **ribs** **pelvis** **skull** **thigh bone (femur)**



1 _____ 2 _____ 3 _____

4 _____ 5 _____ 6 _____

2 Draw a line to match. [4]

Draw a line to match each pair, or write the letter.

skull ●	● **A** protects the brain
ribs ●	● **B** holds the body up
backbone ●	● **C** protect the heart and lungs
muscles ●	● **D** pull bones to move them

3 **True or false?** [3]

Statement	True	False	If false, correct it
1. Muscles can push and pull.			
2. Adults have 206 bones.			
3. It is safe to share medicine with a friend.			

How do we move?

★★☆
Grade 3

Name: Class: Date:

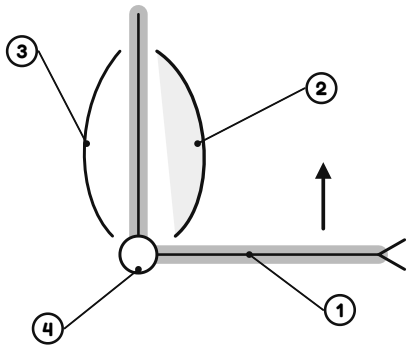
Learning goals

- Explain muscle pairs.

1 Label the arm muscles. [4]

LABELS

triceps (relaxes) bone elbow joint biceps (contracts)



1 _____ 2 _____

3 _____ 4 _____

2 What happens to the biceps when you bend your arm? [2]

.....

.....

3 Why do muscles have to work in pairs? [2]

.....

.....

4 Why are some medicines kept in cupboards high up? [1]

.....

INVESTIGATE · A1 · SKELETONS AND MEDICINES

Do longer arms throw further?

★★☆ Grade 3

Name: Class: Date:

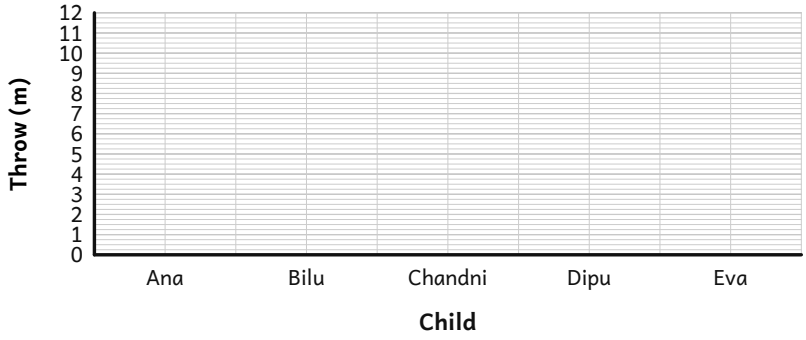
Learning goals

- Look for a pattern.
- Measure length.

Class 3B measured the arm length of five children and how far each could throw a bean bag (best of three throws).

1 **Draw a bar chart.** [3]

Child	Ana	Bilu	Chandni	Dipu	Eva
Arm length (cm)	48	52	50	55	46
Throw (m)	8	10	7	11	9



2 **Did the child with the longest arms throw furthest?** [1]

3 **Is there a clear pattern? Explain.** [2]

REAL-WORLD CASE STUDY · A1 · SKELETONS AND MEDICINES

Strong bones from small fish

★★★ Grade 3

Name: **Class:** **Date:**

Learning goals

- Link food to strong bones.
- Use numbers.

Science · Health · Food · Maths

Bones need **calcium** to grow strong. Milk, yoghurt (*doi*) and green leafy vegetables have calcium. In Bangladesh, tiny fish such as **mola** and **dhela** are eaten whole, with their bones, so they give lots of calcium too.

Scientists and nutrition workers in Bangladesh have encouraged farmers to grow mola fish in their ponds, so that children get more calcium and vitamins.

Based on nutrition research about small fish in Bangladesh.

1 **Circle the foods that help build strong bones.** [3]



2 **Why are small fish eaten whole so good for bones?** [2]

3 **A child needs 3 glasses of milk a day. How many in a week?** [1]

Show what you know

★★★
Grade 3

Name: Class: Date:

Learning goals

- Use body words.

1 Find the words.

R	E	P	M	U	S	C	L	E
I	Y	B	W	S	K	U	L	L
B	T	Y	M	H	U	S	L	B
S	P	H	I	M	B	O	W	O
M	E	D	I	C	I	N	E	N
D	Y	W	W	Y	N	D	D	E
A	B	U	N	V	S	J	S	D
A	H	N	R	O	S	U	G	L
R	E	K	Y	O	P	U	L	L

bone medicine muscle pull ribs skull

2 Which bones protect the heart? [1]

◯ skull ◯ ribs

◯ leg bones

3 When a muscle contracts, it... [1]

◯ gets longer ◯ gets shorter and pulls

◯ pushes

4 Challenge. Animals like worms and jellyfish have no bones. How are their bodies different from ours? [2]

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
say what the skeleton does	○	○	○
name important bones	○	○	○
explain how muscles work in pairs	○	○	○
use medicines safely	○	○	○
name foods for strong bones	○	○	○